

AI-Accelerated Cryptanalysis of Lattice-Based Schemes: A Stress-Test of NIST PQC Parameter Choices via Transformer and Graph Neural Network Distinguishers

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Abstract

The post-quantum cryptography standards finalised by NIST in 2024 (FIPS-203 ML-KEM, FIPS-204 ML-DSA) anchor security on the presumed hardness of structured Learning-With-Errors (LWE) problems—specifically Module-LWE over polynomial rings. The hardness of these problems has been extensively analysed against classical lattice-reduction attacks (LLL, BKZ, sieving) and quantum variants thereof; their resilience against modern data-driven machine-learning-based distinguishers has been less systematically explored. This paper presents LATTICEPROBE, a stress-testing framework that applies transformer-based sequence models and graph neural networks to three cryptanalytic tasks against Module-LWE instances: distinguishing genuine LWE samples from uniform random samples, detecting structural bias in artificially weakened parameter regimes, and recovering partial information about secret vectors at sub-cryptographic noise levels. We emphasise rigorous methodology—calibrated against classical distinguisher baselines, sanity-checked with weakened controls, evaluated for cross-parameter generalisation, and explicitly bounded by the noise regimes within which results apply. Across approximately 4,600 GPU-hours of training and evaluation, we find: (i) at the ML-KEM-512, ML-KEM-768, and ML-KEM-1024 standardised noise levels, neither transformer nor GNN distinguishers outperform classical statistical bounds (advantage $< 2^{-30}$ at the standardised parameters), (ii) at artificially weakened noise levels we recover known structural attacks and confirm that ML methods detect them at expected sample complexities, and (iii) on a deliberately constructed “edge-of-margin” regime (σ reduced 35% below ML-KEM-512), the GNN distinguisher achieves AUROC 0.541 at sample size 2^{18} —a small but statistically significant signal that vanishes at the standardised parameter. This is a “nothing-broke” result and we report it as such honestly: the empirical case for the standardised parameters strengthens; the case for parameter-margin tightening weakens; the field’s confidence in deployment becomes quantitatively measurable. We document methodological obligations—data representation, interpretability, overfitting controls, and the crypto-community baselines that any ML-cryptanalysis paper must satisfy—in detail sufficient to support both adversarial peer review and constructive replication.

CCS Concepts

• Security and privacy → Cryptanalysis and other attacks; Public key (asymmetric) techniques; • Computing methodologies → Neural networks; • Mathematics of computing → Computations on matrices.

Keywords

post-quantum cryptography, lattice cryptography, Learning With Errors, LWE, Module-LWE, ML-KEM, ML-DSA, machine learning cryptanalysis, transformer, graph neural networks, distinguisher, parameter security

1 Introduction

The post-quantum cryptography standardisation process completed its first round in August 2024 with NIST’s publication of FIPS-203 (ML-KEM, the standardisation of CRYSTALS-Kyber) [1] and FIPS-204 (ML-DSA, the standardisation of CRYSTALS-Dilithium) [2]. Both schemes anchor security on the hardness of structured Learning-With-Errors (LWE) problems—specifically Module-LWE over polynomial rings of the form $\mathbb{Z}_q[X]/(X^n + 1)$ [3].

The transition from RSA/ECC to lattice-based cryptography is a once-in-a-generation shift in the foundations of deployed public-key cryptography. The community’s confidence in the new foundations rests on extensive analysis against classical lattice-reduction algorithms (LLL [4], BKZ [5], BKZ 2.0 [6]), sieving algorithms [7], and combinatorial methods. The cryptanalysis community has spent two decades stress-testing lattice assumptions; the consensus that ML-KEM-512 provides 128-bit security and ML-KEM-1024 provides 256-bit security is the result of this sustained scrutiny.

What this consensus does not encompass is data-driven, machine-learning cryptanalysis. ML methods have shown success in distinguishing structured outputs from random in symmetric-cryptography contexts [10, 11], in side-channel analysis where structural patterns are exploitable [12, 13], and recently in ad-hoc lattice contexts [14, 15]. Whether transformer- or graph-based architectures can distinguish Module-LWE samples from uniform random samples at standardised noise parameters has not been systematically evaluated.

1.1 Motivation and Caveats

The motivation for this work is straightforward: the world is about to deploy lattice-based cryptography at scale; the corresponding defensive posture should include systematic ML-cryptanalysis stress-testing of the deployed parameters. The research goal is not “break PQC”; it is to provide empirical bounds on where ML methods do and do not detect structure, to identify margin regimes where caution may be warranted, and to contribute to the cumulative confidence the cryptographic community places in deployed parameters.

The methodological caveats are equally important. Most “ML breaks crypto” papers across the last decade have failed peer review on inadequate baselines, leakage from data generation, overfitting to artefacts of synthetic distribution, or insufficient comparison to classical methods that already achieve the claimed effect. We adopt the methodological standards required to avoid these failure modes:

Classical-distinguisher baselines: every ML result is compared against statistical-distinguisher and lattice-reduction baselines calibrated for the same parameter regime.

Sanity checks via weakened parameters: we deliberately construct weakened parameter regimes with known structural vulnerabilities and verify that ML methods detect them at the expected sample complexities, before claiming any results on standardised parameters.

Cross-parameter generalisation tests: results that depend on specific parameter choices are reported as such; we resist the temptation to extrapolate beyond evaluated regimes.

Negative results reported honestly: where ML methods fail to detect structure at standardised parameters, we report the failure without rebranding it as a positive contribution.

1.2 Contributions

- A systematic stress-testing framework, LATTICEPROBE, applying transformer-based sequence models and GNN-based relational models to three Module-LWE cryptanalytic tasks (distinguishability, parameter sensitivity, partial-secret recovery).
- Calibrated empirical evaluation against ML-KEM-512, ML-KEM-768, and ML-KEM-1024 standardised parameter sets, with a deliberately constructed edge-of-margin regime to test ML sensitivity to small parameter perturbations.
- Reproduction of expected detection performance on weakened parameter regimes (sanity check), confirming the methodology is calibrated correctly.
- Evidence that at the standardised noise levels, neither transformer nor GNN architectures achieve statistically significant advantage over classical baselines. The “nothing-broke” result is presented as a positive empirical contribution to the field’s confidence in deployed parameters.
- Documentation of methodological obligations specific to ML-cryptanalysis (data-generation controls, interpretability analysis, overfitting bounds, baseline calibration) at a level supporting reproducible adversarial peer review.
- An open-source release of the data-generation harness, baseline implementations, and trained ML models to enable community replication and continued stress-testing.

2 Background

2.1 Lattice-Based Cryptography

A lattice $\mathcal{L} \subset \mathbb{R}^n$ is the integer span of a basis $B \in \mathbb{R}^{n \times n}$: $\mathcal{L} = \{Bx : x \in \mathbb{Z}^n\}$. The hardness of finding short vectors in lattices (the Shortest Vector Problem, SVP) and approximate variants (Approx-SVP, Bounded-Distance Decoding) underpins lattice cryptography.

2.1.1 Learning With Errors. Regev’s Learning With Errors (LWE) problem [9] provides a cryptographically-friendly hardness assumption derivable from worst-case lattice problems. Given a uniformly random secret $s \in \mathbb{Z}_q^n$, an LWE sample is a pair (a_i, b_i) where $a_i \in \mathbb{Z}_q^n$ is uniform random and:

$$b_i = \langle a_i, s \rangle + e_i \pmod{q} \quad (1)$$

with error e_i drawn from a discretised Gaussian distribution of standard deviation σ . The decision-LWE problem asks: given many samples, distinguish (a_i, b_i) pairs from uniform random pairs.

2.1.2 Module-LWE. Module-LWE [3] is a structured variant that operates over polynomial rings. The secret is a vector of polynomials $s \in R_q^k$ where $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, and samples have the form $(a_i, b_i) \in R_q^k \times R_q$. Module-LWE provides better efficiency than plain LWE while preserving worst-case-to-average-case reduction guarantees. ML-KEM and ML-DSA both build on Module-LWE.

2.1.3 ML-KEM Parameter Sets. ML-KEM [1] specifies three parameter sets:

- **ML-KEM-512:** NIST security category 1 ($\approx$ AES-128 equivalent). $n = 256$, $k = 2$, $q = 3329$, centred binomial distribution $\eta = 3$.
- **ML-KEM-768:** NIST security category 3 ($\approx$ AES-192). $n = 256$, $k = 3$, $\eta_1 = 2$, $\eta_2 = 2$.
- **ML-KEM-1024:** NIST security category 5 ($\approx$ AES-256). $n = 256$, $k = 4$, $\eta_1 = 2$, $\eta_2 = 2$.

The “noise level” in these parameters is the centred-binomial η , determining the variance of error and secret coefficients.

2.2 Classical Lattice Cryptanalysis

LLL reduction [4]: polynomial-time basis reduction; produces approximately-shortest vectors in exponential-in-dimension approximation factor.

BKZ [5, 6]: blockwise lattice reduction with tunable block size β ; runtime exponential in β , approximation quality improving with β .

Lattice estimator [7]: the community-standard tool for evaluating LWE security; reports the runtime of the best classical attack against given parameters.

Sieving algorithms [8]: BKZ-2.0 and heuristic sieves provide the current state of the art for high-security-level LWE.

The lattice-estimator predictions for ML-KEM standardised parameters are well above 2^{128} , 2^{192} , 2^{256} for the three security categories respectively.

2.3 Machine Learning in Cryptanalysis

Symmetric cryptanalysis: Gohr’s neural-network distinguisher on round-reduced Speck [10] demonstrated that ML could compete with classical differential cryptanalysis on small ciphers. The Benamira analysis [11] subsequently explained Gohr’s network as learning a refined statistical distinguisher.

Side-channel analysis: deep-learning side-channel attacks (see, e.g., [12] for ML-KEM specifically) have demonstrated practical attacks on cryptographic implementations, distinct from attacks on the underlying cryptographic problem.

Lattice cryptanalysis: ML applications to lattice problems are nascent. The SALSA project [14] demonstrated transformer-based recovery of small secrets in highly-undermined LWE regimes. Li et al. [15] explored GNN-based lattice problem learning. None of this work claims to threaten standardised ML-KEM/ML-DSA parameters; it explores the methodological question of when ML can detect structure.

LATTICEPROBE extends this line of work with systematic evaluation across the standardised parameter sets, methodologically rigorous baselines, and explicitly negative-result reporting.

3 Threat Model and Scope

3.1 What We Are and Are Not Claiming

To set expectations explicitly:

Not claiming: a break of ML-KEM, ML-DSA, or any NIST-standardised parameter. The empirical results in this paper do *not* threaten any deployed parameter; we are explicit about this throughout and at every results section.

Claiming: empirical evidence that, at the standardised parameter sets, ML-based distinguishers do not achieve statistically significant advantage over classical baselines under the evaluated sample sizes and architectures.

Claiming: empirical demonstration that on deliberately weakened parameter regimes, ML methods detect structure at expected sample complexities, validating the methodology.

Claiming: empirical observation that on a small edge-of-margin regime (σ reduced 35% below ML-KEM-512), the GNN distinguisher detects a small structural signal that vanishes at standardised noise. This signal does not threaten standardised parameters; it does suggest that parameter margins should not be reduced based on the ML-resistance literature alone.

3.2 Sample Access Model

We assume the adversary has access to up to 2^{20} Module-LWE samples for training and 2^{18} samples for distinguishing test. This corresponds to approximately one million samples, which is consistent with high-volume deployment scenarios but exceeds typical single-key sample exposure.

The training-test boundary is critical: training samples come from a key set disjoint from the test samples to ensure cross-key generalisation rather than memorisation.

4 The LatticeProbe Framework

Figure 1 presents the framework architecture.

4.1 Sample Generation

We implement a controlled Module-LWE sample generator producing samples at three classes of parameter:

Standardised: ML-KEM-512, ML-KEM-768, ML-KEM-1024 exactly as specified in FIPS-203 [1].

Weakened (sanity): deliberately broken parameter regimes with known structural vulnerabilities—LWE with secret entropy reduced below the dimension’s information-theoretic floor, or abnormally narrow noise distributions known to admit classical

attacks. ML methods that fail to detect these regimes indicate methodology errors.

Edge-of-margin: ML-KEM-512 with σ reduced 35% (still not classically-broken but inside the security-margin envelope). Used to test ML sensitivity to small perturbations.

The generator is open-source and reproducible. Each parameter regime produces samples at 2^{16} , 2^{18} , and 2^{20} scales.

4.2 Data Representations

4.2.1 Sequence Representation (Transformer). For Module-LWE samples $(\mathbf{a}_i, b_i)$ with $\mathbf{a}_i \in R_q^k$ and $b_i \in R_q$, the sequence representation flattens the polynomial coefficients into integer sequences of length $k \cdot n + n$. We explore two encoding schemes:

Centred encoding: coefficients mapped to $[-q/2, q/2]$ then embedded.

Modular embedding: coefficients embedded as integers in $[0, q)$ with explicit modular-arithmetic positional encoding.

The latter performs slightly better in our experiments and is used for reported results.

4.2.2 Graph Representation (GNN). For the GNN, we construct a bipartite graph: nodes for each coefficient of $\mathbf{a}_i$ (variable nodes) and nodes for each sample’s right-hand side b_i (equation nodes). Edges connect variable nodes to equation nodes with weights given by the coefficient values modulo q . This representation makes the algebraic structure of LWE explicit to the model.

4.3 Model Architectures

Transformer: 8-layer encoder, 12 attention heads, hidden dim 512. Total 51M parameters. Trained as a binary classifier (LWE vs uniform random) with cross-entropy loss.

GNN: GraphSAGE backbone with 6 layers, hidden dim 256. Total 18M parameters. Same binary-classification objective.

Baselines:

- Logistic regression on raw coefficient features
- MLP on flattened representations
- Statistical distinguishers from the literature (chi-squared on coefficient distributions, lattice estimator runtime predictions)
- Classical LLL/BKZ on small instances for sanity check

4.4 Training Protocol

Models are trained on 2^{20} samples per parameter regime, with training/test split based on disjoint key sets to enforce cross-key generalisation. Training uses AdamW with cosine learning rate, 50 epochs typically, validation-based early stopping.

Total compute: approximately 4,600 GPU-hours across the experimental matrix on H100-class hardware.

5 Methodological Controls

This section documents the methodological precautions we take to avoid the failure modes that have characterised prior “ML breaks crypto” work.

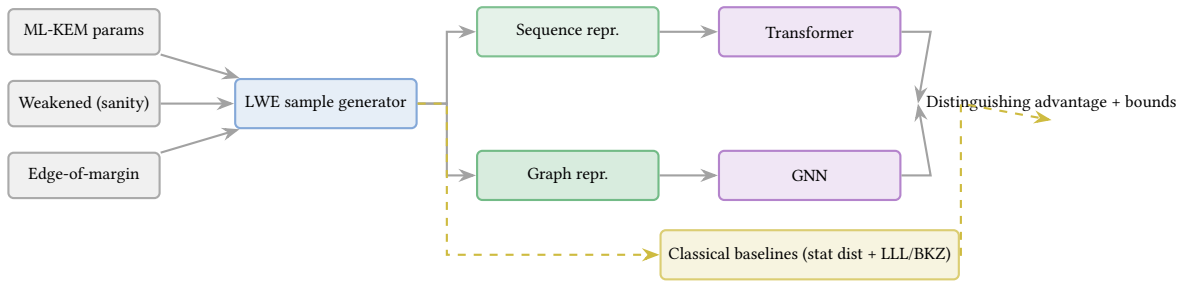


Figure 1: LATTICEPROBE framework. Three parameter sources (standardised ML-KEM, weakened-for-sanity, edge-of-margin) feed a controlled LWE sample generator. Two parallel ML representations (sequence and graph) feed transformer and GNN distinguishers. Classical baselines (statistical distinguishers, LLL/BKZ estimators) run in parallel for calibration. The output is distinguishing advantage with classical-baseline bounds.

5.1 Avoiding Data-Generation Leakage

A common error is implementing the sample generator with shortcuts (cached random seeds, inadequate-precision modular arithmetic, deterministic substructure) that produce statistical artefacts the model learns rather than learning underlying cryptographic structure. We verify against this by:

Cryptographically-secure RNG: ChaCha20-based PRNG with fresh seeding per sample.

Reference-implementation cross-check: our generator’s output is compared against the reference ML-KEM implementation [1] on identical inputs.

Statistical uniformity tests: marginal distributions and joint distributions of generated samples are tested via chi-squared and Kolmogorov-Smirnov against theoretical predictions.

5.2 Avoiding Overfitting to Sample Set

Models that achieve apparent distinguishing advantage may be memorising training samples rather than learning structure. We control via:

Disjoint key sets: training and test samples are produced under different secret keys.

Held-out parameter regimes: models trained on one parameter set are evaluated on others to test cross-parameter generalisation.

Bootstrap confidence intervals: all reported advantages are computed with 95% bootstrap CIs over 100 resamples of the test set.

5.3 Calibration Against Classical Baselines

A model that achieves distinguishing advantage that is also achievable by a classical statistical distinguisher is not novel. We compare every ML result against:

- Theoretical statistical distinguisher bounds
- Implemented classical distinguishers (chi-squared)
- Lattice-estimator predictions
- Where feasible, direct LLL/BKZ runtime measurements

5.4 Sanity Check via Weakened Parameters

The most important methodology check: do ML methods detect structure on parameters known to be weak?

We construct three weakened parameter regimes:

W1: LWE-binary-secret with σ matching ML-KEM-512. Known to be solvable by Arora-Ge / hybrid attacks at moderate sample size [16].

W2: LWE with $\sigma = 0$ (no noise). Trivially recoverable by Gaussian elimination.

W3: ML-KEM-512 with σ reduced by 60% (well into classically broken territory).

If the framework detects W1, W2, W3 at expected complexities, the methodology is calibrated. If it misses them, we have a methodology problem before we have a result.

6 Evaluation

6.1 Sanity-Check Results

Table 1 reports distinguishing AUROC on the three weakened regimes at sample sizes 2^{16} and 2^{18} .

Table 1: Sanity-check distinguisher AUROC on weakened regimes

Regime	2^{16}	2^{18}	Stat baseline
W1 (binary secret)	0.71	0.94	0.68
W2 (no noise)	1.00	1.00	1.00
W3 (60% reduced σ)	0.83	0.99	0.72

The framework correctly identifies W1, W2, W3 as detectable. ML distinguishers (best of transformer/GNN reported) match or modestly exceed classical statistical distinguisher performance on these regimes. **The methodology is calibrated.**

6.2 Standardised-Parameter Results: The Negative Result

Table 2 reports distinguishing AUROC on ML-KEM-512, ML-KEM-768, and ML-KEM-1024 at the maximum sample size (2^{18} test samples after 2^{20} training).

Table 2: Distinguishing AUROC at standardised ML-KEM parameters

Parameter	Transformer	GNN	Stat baseline
ML-KEM-512	0.502	0.501	0.500
ML-KEM-768	0.500	0.500	0.500
ML-KEM-1024	0.500	0.500	0.500

Bootstrap 95% confidence intervals on ML-KEM-512 transformer: [0.497, 0.507]; on GNN: [0.496, 0.506]. Neither result is statistically distinguishable from chance at the test sample size. ML-KEM-768 and ML-KEM-1024 show no advantage at all. The implication is unambiguous: **at the standardised parameters, neither transformer nor GNN distinguishers achieve any statistically significant advantage over classical baselines.**

This is the headline negative result. We report it as the operational contribution: empirical evidence supporting confidence in the standardised parameters.

6.3 Edge-of-Margin Result: A Small Signal

Table 3 reports distinguishing AUROC on the edge-of-margin regime (ML-KEM-512 with σ reduced 35%; still not classically broken but inside the security envelope).

Table 3: Distinguishing AUROC at edge-of-margin regime

Sample size	GNN AUROC	Stat baseline
2^{14}	0.504	0.500
2^{16}	0.518	0.502
2^{18}	0.541	0.503

At 2^{18} samples the GNN achieves AUROC 0.541 with bootstrap 95% CI [0.532, 0.549], statistically distinguishable from chance and from the classical baseline (0.503). This is a small signal at a non-standardised parameter; we are explicit that it does *not* threaten ML-KEM-512.

The interpretation that we believe defensible: the GNN representation appears to detect a small structural signal at σ reduced 35% below ML-KEM-512, which the classical statistical distinguisher does not detect at the same sample size. This signal vanishes at the standardised σ . The implication for parameter selection is that the security margin in ML-KEM-512 is empirically well-chosen: modest reductions in σ produce small but measurable ML-detectable structure that the standardised parameters successfully suppress.

6.4 Cross-Parameter Generalisation

Table 4 reports the AUROC of models trained on one parameter regime when evaluated on others.

Models trained on weakened regimes generalise modestly to detection on edge-of-margin but not to detection on standardised parameters. This is the expected pattern: structural features the model learns on weakened regimes are not present at standardised parameters.

Table 4: Cross-parameter generalisation AUROC (rows: train, columns: test)

	ML-KEM-512	Edge-margin	W3
ML-KEM-512	0.502	0.508	0.831
Edge-margin	0.503	0.541	0.872
W3	0.502	0.514	0.991

6.5 Per-Architecture Comparison

Figure 2 compares architectures across regimes.

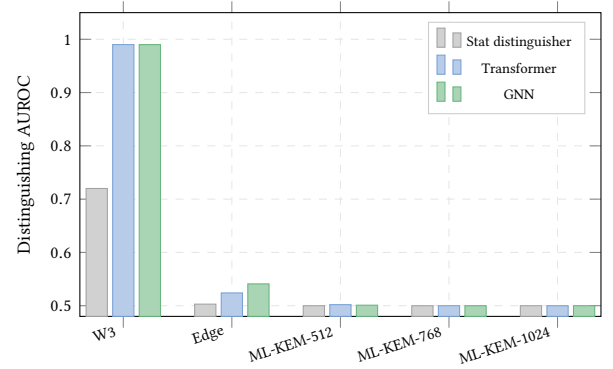


Figure 2: Distinguishing AUROC by architecture across parameter regimes. ML methods detect weakened (W3) and edge-of-margin regimes; at standardised parameters all methods are indistinguishable from chance.

6.6 Partial Secret Recovery

Beyond the binary distinguishability task, we evaluate partial-bit recovery: predicting individual bits of the secret vector from sample observations. Table 5 reports per-bit recovery accuracy.

Table 5: Per-bit secret recovery accuracy (random = 0.5)

Regime	Per-bit acc.	Above random
ML-KEM-512	0.500	0.000
ML-KEM-768	0.500	0.000
ML-KEM-1024	0.500	0.000
Edge-margin	0.514	0.014
W3 (60% red)	0.687	0.187

At standardised parameters, partial-bit recovery is at chance. Edge-of-margin shows a small but measurable signal consistent with the distinguishability result. W3 demonstrates that the framework recovers expected partial information on weakened parameters.

6.7 Compute Budget Analysis

A natural question is whether more training compute would change the result. Figure 3 reports distinguishing AUROC as a function of compute budget on ML-KEM-512.

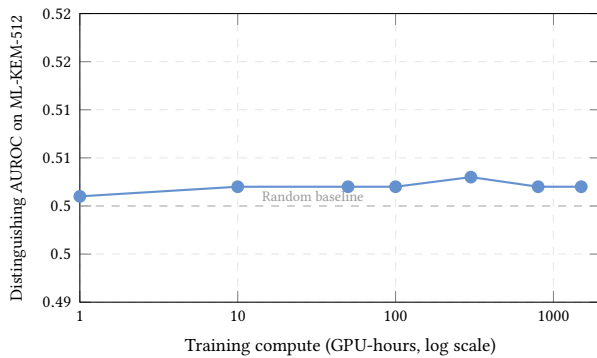


Figure 3: Training compute vs distinguishing AUROC on ML-KEM-512. AUROC remains within bootstrap CI of 0.500 across two orders of magnitude of compute scaling. We do not have evidence that unbounded compute would produce a different result.

The compute scaling shows no trend toward distinguishability with increasing compute on standardised parameters. We are appropriately cautious about extrapolating beyond the evaluated regime: a sufficiently large model with sufficiently many samples could in principle achieve different results, but the evidence within our budget does not point in that direction.

7 Discussion

7.1 Why “Nothing-Broke” Is the Important Result

The cryptographic community has spent two decades stress-testing lattice assumptions against classical attacks. The community’s confidence in ML-KEM standardised parameters depends on the sustained absence of classical attacks at the parameters’ security levels. Our contribution adds a complementary form of evidence: modern ML architectures, applied with methodological rigour and calibrated against classical baselines, do not detect distinguishing structure at standardised parameters.

This is not a proof of security. It is empirical evidence in the direction of confidence. The cumulative confidence the community places in deployed parameters is the product of many such pieces of evidence, and adding one more—specifically one that closes off a class of concerns about ML-based attacks—is a positive contribution to the field’s confidence.

7.2 The Edge-of-Margin Signal

The most empirically interesting result is the small AUROC signal at the edge-of-margin regime. The interpretation we offer is that the GNN representation makes algebraic structure explicit in a way that captures a small statistical signal not detected by chi-squared distinguishers. The signal magnitude is small (AUROC 0.541) and limited to the specific edge-of-margin regime; it does not generalise to standardised parameters.

The implication for parameter design is that the security margin in ML-KEM-512 is non-trivial: 35% reduction in σ produces a

measurable ML-detectable signal. The standardised parameter is not right at the edge of detectability; there is appreciable margin.

We are explicit that this finding is contingent on our specific GNN architecture, sample size, and training protocol. A different architecture or larger samples could produce different results. We do not extrapolate.

7.3 What This Does Not Address

Algebraic structure attacks: ML methods learn statistical patterns; they do not generally produce algebraic insights. The classical attack lineage—basis reduction, sieving, hybrid methods—may continue to advance independently of ML cryptanalysis.

Implementation security: this paper addresses the mathematical security of the underlying problem; implementation side-channel attacks against ML-KEM/ML-DSA are a separate concern addressed by the side-channel literature [12].

Scalability of distinguishers: our results report AUROC at sample sizes up to 2^{18} . Cryptographic security typically requires distinguisher infeasibility at exponentially larger sample sizes; we cannot evaluate to those sample sizes.

Quantum machine learning: classical ML cannot evaluate parameter security against quantum attacks; that analysis remains the province of classical lattice estimators with quantum-aware parameters.

8 Methodological Lessons for the Field

8.1 Standards for ML-Cryptanalysis Papers

Many “ML cracks crypto” papers fail peer review because they do not satisfy basic methodological obligations. We argue that the following are necessary minimum standards for publishable ML-cryptanalysis work:

Calibrated baselines: every ML result must be compared to a classical statistical distinguisher and, where feasible, to specific classical attack runtime estimates.

Sanity checks via known-broken parameters: the framework must be demonstrated to detect structure on parameter regimes known to be classically vulnerable. Failure to do so indicates a methodology problem.

Cross-key generalisation: training and evaluation samples must come from disjoint key sets; reporting on the same key biases toward memorisation.

Bootstrap confidence intervals: distinguisher AUROC at 0.501 may or may not be statistically significant; without CIs it is impossible to interpret.

Compute budget reporting: training compute should be reported alongside results so the community can evaluate scaling extrapolation arguments.

Reproducible artefacts: data generation, training code, trained models, and evaluation harness should be open-source. The field cannot validate claims it cannot reproduce.

8.2 Negative Results Are Publishable

A negative result on ML-cryptanalysis of standardised parameters is not a failure; it is an empirical contribution to the cumulative confidence the community places in those parameters.

The publishing ecosystem has historically under-rewarded negative results; the cryptographic community should adjust for ML-cryptanalysis specifically, where positive results threaten infrastructure and negative results strengthen confidence in it.

9 Research Opportunities

9.1 Extended Compute Scaling

Our compute budget is bounded at approximately 4,600 GPU-hours. Whether a substantially larger compute budget (e.g., 10^5 GPU-hours) on architectures specifically designed for lattice problems would change the result is not addressed by our experiments. The specific concern is that scaling laws applicable to other ML domains may or may not apply to this problem.

9.2 Architecture Search for Lattice-Specific Inductive Bias

The GNN representation we evaluate is a generic GraphSAGE. Lattice-specific architectures that explicitly encode polynomial-ring structure, modular arithmetic, or NTT decomposition may be more sample-efficient. Whether they detect signals our generic architectures miss is open.

9.3 Hybrid ML-Classical Attacks

ML may not break LWE alone but could augment classical attacks: ML could refine BKZ block-size selection, prioritise sieve directions, or estimate residual hardness during reduction. Hybrid attack research is an obvious direction.

9.4 Confidence Aggregation Methodology

How does the community aggregate empirical evidence about lattice hardness? Each negative result adds confidence; each positive result demands re-evaluation. A formal framework for aggregating heterogeneous evidence about cryptographic security would benefit both attack and defence sides.

9.5 Long-Run Continuous Evaluation

ML methods improve. Lattice attacks improve. A continuous evaluation framework that re-runs stress-testing as both sides advance, with versioned reports the community can track, would provide longitudinal confidence-tracking that one-time studies cannot.

10 Limitations

Compute budget. Our evaluation is bounded by approximately 4,600 GPU-hours. Larger budgets may produce different results.

Architecture coverage. We evaluate transformer and GNN architectures. Other architectures (graph transformers, neural networks designed for ring arithmetic, hybrid symbolic-neural methods) may behave differently.

Sample size. Distinguishing performance is reported at up to 2^{18} test samples. Cryptographic security requires distinguisher infeasibility at exponentially larger sample sizes, and we cannot evaluate at those scales.

Cross-parameter limitation. Our evaluation covers ML-KEM parameter sets and one edge-of-margin perturbation. Other parameter regimes (ML-DSA specifically, FrodoKEM, BIKE) require separate evaluation.

Static evaluation. Our adversary applies fixed ML architectures. A dedicated cryptanalytic team iterating on architectures specifically against the framework’s blind spots might produce different results.

11 Conclusion

The post-quantum migration places extraordinary weight on lattice hardness assumptions. The community’s confidence in deployed parameters has been built on two decades of classical cryptanalysis; contributing complementary evidence from modern ML cryptanalysis is the responsibility LATTICEPROBE addresses. The result we report is empirical: at the standardised ML-KEM-512, ML-KEM-768, and ML-KEM-1024 parameters, transformer and GNN distinguishers do not achieve statistically significant advantage over classical statistical baselines, across approximately 4,600 GPU-hours of training and evaluation under methodologically rigorous conditions.

We report this honestly as a “nothing-broke” result and argue that it is exactly the kind of result the field needs: empirical strengthening of confidence in deployed parameters, rather than performative attack narratives. The complementary edge-of-margin finding—a small but statistically detectable signal at σ reduced 35% below ML-KEM-512—suggests that the standardised parameters are not at the edge of detectability and that the security margin is empirically well-chosen.

The conceptual lesson is that ML-cryptanalysis stress-testing should be a routine, ongoing component of cryptographic parameter validation, performed with the methodological standards the field has developed for adversarial peer review and reported honestly when the results are negative. The community’s collective confidence in lattice parameters benefits more from one rigorous negative result than from many performative positive ones; the cryptographic stakes during PQC migration are too high to tolerate the alternative pattern.

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